

Will R. Boone

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PROFESSIONAL SUMMARY

Computer Science student at the University of Arkansas (GPA: 3.84) with hands-on experience in network security tooling, distributed systems, and backend engineering. Built a real-time network intrusion detection system in Go with protocol-level packet parsing, ARP spoofing detection, and DNS response anomaly detection, and a Go-based malware analysis sandbox combining static binary analysis with dynamic execution monitoring. Seeking software engineering roles in cybersecurity and defense where low-level systems, network programming, and reliable backend design matter.

EDUCATION

University of Arkansas

Fayetteville, AR

Bachelor of Science in Computer Science. Chancellor's and Dean's List: Fall 2023, Spring 2025, & Spring 2026 | New Arkansan Non-Resident Tuition Award Scholarship. GPA: 3.84 / 4.00. Expected May 2027

Relevant Coursework: Software Engineering, Database Management Systems, Programming Paradigms, Artificial Intelligence, Programming Foundations I & II

EXPERIENCE

Volunteer

Summer 2022 – Summer 2024

Lord of the Streets

Houston, TX

- Served Houston's unhoused population through direct community outreach, executing responsibilities independently while upholding organizational safety protocols in a high-variability environment
- Demonstrated adaptability and task prioritization across concurrent responsibilities in a dynamic, unstructured setting

Delivery Driver

Summer 2021 – Summer 2023

Stefano's Pizza

Friendswood, TX

- Managed high-volume delivery routes under tight time constraints, maintaining order accuracy and customer satisfaction through disciplined, independent execution
- Built operational discipline and time management navigating unpredictable, high-pressure conditions with minimal oversight

PROJECTS

go-sentinel: Network Intrusion Detection Tool

Summer 2026

Go

- Engineered a real-time network monitor in Go that captures live packets via libpcap and decodes Ethernet, IPv4, TCP, UDP, and DNS headers from raw bytes using hand-written parsers, tracking per-flow statistics across a concurrent, mutex-guarded state model
- Implemented detection heuristics for TCP port scans and SYN-flood attacks using rolling time-windows and configurable thresholds, with per-source alerting and de-duplication to suppress repeat alarms

GoSandbox: Malware Analysis Sandbox

Summer 2026

Go

- Built a malware analysis sandbox in Go combining hand-written PE/ELF binary parsers with dynamic execution monitoring, correlating static and behavioral signals into a weighted 0–100 threat score with MITRE ATT&CK-mapped findings
- Implemented sandboxed execution with cgroup/ulimit resource confinement, full syscall tracing via strace, and live network capture with DNS/TLS SNI/HTTP extraction, generating self-contained JSON and HTML threat reports

Distributed Task Scheduler

Summer 2026

Python, FastAPI, SQLAlchemy, JavaScript

- Built a distributed job-queue system in which multiple worker processes poll a FastAPI backend to claim and execute priority-ordered tasks, with atomic job assignment designed to prevent duplicate claims under concurrent load; validated with 3-4 workers draining 25+ jobs in under 3 minutes with no duplicate assignments observed
- Implemented automatic retry with configurable per-job limits, dead-worker detection with job requeueing, and a live dashboard (2-second polling) surfacing throughput, failure rate, average duration, and a per-worker efficiency leaderboard

SKILLS

Languages: Go, Python, C++, Java, JavaScript, SQL, HTML/CSS

Tools & Technologies: libpcap, YARA, MITRE ATT&CK, FastAPI, SQLAlchemy, Git

Human Languages: Spanish (Intermediate), Japanese (Upper-Beginner)

ACTIVITIES & INTERESTS

- BYX Fraternity (3-year contributing member), University of Arkansas ACM, Fayetteville Young-Life
- Interests: International Travel, Japanese Language & Culture, Fitness, Camping, Cooking, Faith & Community Service